

HELENA FU

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EDUCATION

Tufts University - School of Engineering

B.S. in Computer Science, B.S. in Mathematics (GPA: 3.93 / 4.00)

Medford, MA

December 2026

Honors: 3x Tufts Dean's List, Al Gardner Memorial Endowment Scholarship (Society of Women Engineers).

Relevant Coursework: Machine Learning Foundations, Linear Algebra, Differential Equations, Data Structures, Machine Structure & Assembly Programming, Mathematical Aspects of Data Analysis, Algorithms, Real Analysis, Mathematical Modeling.

TECHNICAL SKILLS

Languages: Java, Python, C/C++/C#, SQL (Postgres), JavaScript, HTML/CSS, R, Assembly, ROS, Bash Script, MatLab, OCaml, PHP

Frameworks: React, Node.js, Flask, WordPress, Next.JS

Developer Tools: Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse

Libraries/API: pandas, NumPy, Matplotlib, TensorFlow, OpenCV, Keras, PyTorch, sklearn, Claude, Gemini, GPT

EXPERIENCE

Human-Robot Interaction Lab (HRI) at Tufts University

Research Assistant

Feb. 2024 – Present

Medford, MA

- Developed pupillometry signal processing method in **Python** for cognitive load analysis, improving data integrity by **30%**.
- Programmed automated robot behavioral responses to over **20 LLM** commands using **ROS** and **ActionScript**.
- Built a multiplayer Unity VR game integrated with **C#**, **ROSBridge**, and **DIARC**, facilitating studies with **50+** participants.

MIT Schwarzman College of Computing

Machine Learning Intern

Aug. 2024 – Dec. 2024

Cambridge, MA

- Processed and integrated **11GB** of water particle and biochemical data across **3** data sources.
- Developed **multi-modal supervised learning** solution predicting hydrographic measurements with **97%** accuracy.
- Predicted HAB risk with **76%** accuracy and identified **10+** early indicators of harmful algae blooms.
- Selected for Break Through Tech AI @ MIT from **3000+** applicants.

Lawrence Berkeley National Laboratory at University of California, Berkeley

Research Intern

June 2023 – July 2023

Berkeley, CA

- Built **100+** image recognition **convolutional neural networks (CNN)** for heterogeneous hardware computing optimization.
- Classified CIFAR10 dataset with **83%** accuracy utilizing **PyTorch** and **TensorFlow**.

PROJECTS

ML Quantitative Trading | The Herd Project Selection

Jan. 2025 – Present

- Designed automated trading algorithms achieving a **53% increase** in profit compared to moving average benchmark.
- Deployed **RL**, **LSTM**, and **HMM** with real-time monitoring to refine model accuracy and profit potential.

DSCVR.AI | Cornell NYC AI Hackathon Winner

Feb. 2025 – Mar. 2025

- Developed hyper-personalized research and writing environment using **Docker**, **Claude API**, and **Amazon Bedrock AWS**.
- Engineered a seamless UX with **Next.js** and **Tailwind**, earning Top Prize, Best UX, and Most App Store Ready awards.

Image Recognition Dermatology Competition | 6th Place out of 350+

Jan. 2025 – Present

- Designed custom **TorchVision**, **Sklearn** and **TensorFlow** image recognition neural networks achieving a **72% accuracy**.
- Won top award from Algorithmic Justice League for developing innovative **image preprocessing** algorithms.

Government Investment for Life Expectancy | 2nd Place Tufts DataMania Hackathon

Mar. 2024 – Mar. 2024

- Developed **Keras** model correlating government spending with life expectancy achieving **77%** accuracy.
- Created a comprehensive report on optimized funding strategies for public health.